A-9

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year 2nd Semester 2011
Course: PHY 1221 (Applied Electricity and Magnetism)
Time: 4 Hrs. Full Marks: 52.5
[N.B. Answer SIX questions taking at least THREE from each part.]

Part A

1.a) Define an electric dipole and its moment. 1
b) Find the electric field $\vec{E}$ at a point P due to a dipole at a distance r from the midpoint of the dipole. 5
c) A water molecule in its vapour state has an electric dipole moment 6.2×10^{-30}cm. What is the electric field $\vec{E}$ at a distance r of 1.1 nm from the molecule on the dipole axis? 2.75

2.a) What is the role of dielectric constant in a capacitor. 1.75
b) Explain susceptibility and electric polarization. Show that $\vec{D} = \epsilon_0\vec{E} + \rho$. 3+4

3.a) Define current density and conductivity. 2
b) Establish the relation between current density and electron drift velocity. 3
c) One end of an Al wire of diameter 2.5 mm is welded to one end of Cu wire of diameter 1.8 mm. The composite wire carries a steady current of 1.3A. What is the current density in each wire? 3.75

4.a) Describe Kirchhoff's laws in a network. 3
b) Deduce the condition of balance of Wheatstone's bridge. 5.75

Part B

5.a) State and explain Faraday's law of induction. 1.75
b) Show that the mutual inductances are complementary. 2
c) Find the co-efficient of coupling. 2
d) A 0.5 m long solenoid has 1500 turns and area of cross section 0.0075 m^2. A second coil of 500 turns is closely wound on the central part of the solenoid. Calculate mutual inductances between two coils. 3

6.a) Explain Peltier and Thomson effects. 3
b) How the Peltier effect differs from Joule heating effect? 3
c) The emf in a Fe-Pb thermocouple, the cold junction of which is kept at 0°C is given by $E=18t-0.015t^2$. Find the neutral temperature and Peltier co-efficient at 27°C. 2.75

7.a) Find expression for the growth and decay of charges of a capacitor through a resistor with constant emf. 3.75
b) What is the time constant in an RC circuit? Show that the time constant has the unit of time. 2
c) In an RC circuit $R=0.4\times10^6$ ohm and $C=2.5\times10^6$ F, in what time will the charge in the capacitor decay one fourth of its initial value? 3

8.a) Find the mean value and rms value of an alternating current. 3
b) Obtain the circuit equation for LR ac series circuit and solve it for the current i. 3.75
c) Define reactance and impedance of the above circuit. Show graphically the variation of reactance with frequency. 2

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engg.(CSE) 1st Year Even Semester 2012

Course: PHY 1221 (Applied Electricity and Magnetism)

Time: 4 Hrs. **Full Marks: 52.5**

[N.B. Answer SIX questions taking at least THREE from each part.]

Part A

1.a) State Gauss's law in electrostatics. 1.75

b) Find an expression for the magnitude of the electric field $\vec{E}$ at a distance r in front of a sheet having surface charge density σ. 4

c) The nucleus of a uranium atom has a radius R of 6.8×10^{-15}m. What is the magnitude of the electric field at its surface? 3

2.a) Define polarization. 2

b) Prove the relation for the capacitance of a parallel plate capacitor $C = \frac{\varepsilon_0 A}{d - t + t/k}$ of plate area A separated by a distance d with dielectric slab of thickness 't' and constant 'k; in between plates. 3.75

c) Derive an expression for the energy stored in a capacitor. 3

3.a) Define resistance, resistivity and conductivity. 2.75

b) What do you mean by a superconductor? 3

c) A Cu wire of diameter 1.8 mm carries a steady current of 2.54A. What is the current density in the Cu wire? 3

4.a) State and Explain Kirchhoff's laws in a network and apply them to find an expression for the current through the galvanometer in an unbalanced Wheatstone bridge. 2+3

b) How can you use this bridge to determine an unknown resistance? 2

c) The current in a simple series circuit is 5 A. When an additional resistance of 2 ohm is inserted, the current drops to 4 A. What was the resistance of the original circuit? 1.75

Part B

5.a) State and prove Ampere's law. Apply it to calculate the magnetic field due to a solenoid. 3+3

b) Explain and find motional emf. 2.75

6.a) Explain Seebeck and Peltier effects. 4

b) What is neutral temperature and temperature of inversion? 2

c) Calculate the maximum emf in an Fe-Pb thermocouple, the cold junction of which is kept at 0°C. Given $a = 13.8 \mu V/^\circ C$; $b = -0.015 \mu V/(^\circ C)^2$ 2.75

7.a) Derive expressions for the growth and decay of current in a circuit having inductance and resistance with constant emf. 3+3

b) An emf 10 volts is applied to a circuit having a resistance of 10 ohm and an inductance of 0.5 henry. Find the time required by the current to attain 63.2% of its initial value. What is the time constant of the circuit? 2.75

8.a) Define reactance and impedance of an ac circuit. 2

b) An alternating emf $E=E_0 \sin \omega t$ is applied to a circuit containing a capacitor C and a resistor R. Find the the current at any instant. 4

c) Explain Resonance in LCR ac series circuit. 2.75

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.) Examination-2013, Year-I, Semester-II

Course: PHY-1221 (Applied Electricity and Magnetism)

Full Marks-52.5 Time: 3 hours

[N.B. Answer any six questions taking THREE from each of the groups]

Part-A

1. a) Derive an expression for the electric field at any point due to and electric dipole. 5
   b) Find the potential energy of an electric dipole placed in a uniform external electric field. 3.75

2. a) Define dielectric constant and energy density. 2
   b) Find the expression for the capacitance of a parallel plate capacitor without dielectric. 3.75
   c) Show that $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$. 3

3. a) Derive an expression for the conductivity of metals based on free electron theory of conductivity. 5
   b) A metal rod of length 25cm has resistance $7.5\times10^{-6}\Omega$. The diameter of the rod is 0.40 cm. Calculate the resistivity of the metal. 3.75

4. a) Define current density. Establish the relation between current density and electron drift velocity. 5
   b) One end of an Al wire of diameter 2.5 mm is welded to one end of Cu wire of diameter 1.8 mm. the composite wire carries a steady current of 1.3 A. what is the current density in each wire? 3.75

Part-B

5. a) State and explain Faraday's law of induction. Deduce its differential form. 3.75
   b) Find the mutual inductance of two co-axial coils. 5

6. a) Write down the laws of intermediate metal and intermediate temperature. 5
   b) Describe Thomson effect. 3.75

7. a) Find expressions for the growth of charges of a capacitor through a resistor with constant emf. 5
   b) In an RC circuit $R = 0.4\times10^{6}\Omega$ and $C=2.5\times10^{6}$F, in what time will the charge in the capacitor decay one fourth of its initial value? 3.75

8. Write short note on (any TWO): 8.75
   a) Seebeck effect
   b) Peltier effect
   c) Reactance
   d) Resonance

University of Rajshahi

Department of Computer Science & Engineering

1st Year Second Semester Examination 2014

Course: PHY 1221 (Applied Electricity and Magnetism)

Full Marks: 52.5 Duration: 3 (Three) Hours

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Answer 06(Six) questions taking any 03(Three) questions from each part

Part-A

1. a) State and prove Gauss's law in electrostatics. 5
   b) Charge is distributed uniformly over the surface of a sphere. Use Gauss's law to find the electric field at the points 3.75
      i. Outside the sphere
      ii. On the surface of the sphere

2. a) Define capacitance and capacitor. 2
   b) Deduce the relation $C = \frac{\varepsilon_0 A}{d}$ for a parallel plate capacitor, where the symbols have their usual significance. 4.75
   c) The parallel plate of an air-filled capacitor are 1cm apart. What must be the area A of each plate if capacitance is to be 0.25μF? 2

3. a) Derive an expression for the energy stored by a charged capacitor. 4.75
   b) The capacitance of a parallel plate capacitor is 400pF and its plate are separated by 2mm of air.
      i What will be the energy when it is charged to 1500 volts? 2
      ii What will be the potential difference with the same charge if the plate separation is doubled? 2

4. a) State and explain Kirchhoff's law in an electrical network. 4
   b) Deduce the condition for the balance of a Wheatstone's bridge. 4.75

Part-B

5. a) State Ampere's law. Derive an expression for magnetizing force, B due to a current carrying long straight conductor. 2+2
   b) Calculate the magnetizing force and flux density at a distance of 1 cm from a long straight circular conductor in air carrying a current of 100A. Draw a curve showing the variation of B from the conductor surface outwards if its diameter is 1mm. 4.75

6. a) Describe a moving coil galvanometer. Explain how current can be measured with it. 3+2
   b) What are the differences between ballistic and deadbeat galvanometer? 1
   c) In a ballistic galvanometer 2μA gives a steady deflection of 1 division. The time of free oscillation is 8s and damping is negligible. What quantity of electricity is measured by a kick of 1 division? 2.75

7. a) A series circuit containing an inductor L and a resistor R is connected to a battery. Obtain the expression for the rate of decay of current. 4.75
   b) An e.m.f. 10v is applied to a circuit having a resistance of 10Ω and an inductance of 0.5H. find the time required by the current to attain 63.2% of its final value. What is the time constant of the circuit? 4

8. a) Define reactance and impedance of an AC circuit. 2
   b) Analyze a series LCR circuit and explain the phenomenon of resonance. 3+1
   c) Find the resonance frequency of an LCR series circuit for L=10mH, C=0.02μF and r=20Ω. 2.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-I Even Semester Exam - 2015
Course: PHY1221 (Applied Electricity and Magnetism)
Full Marks: 52.5 Time: 3 Hours

(Answer any 6 questions not taking more than 3 from each Part)

Part A

1.(a)	Define an electric dipole and its moment.	1
(b)	Find the electric field $\overline{E}$ at a point P due to a dipole at a distance r from the midpoint of the dipole.	5
(c)	A water molecule in its vapour state has an electric dipole moment $6.2\times10^{-30}\ cm$. What is the electric field $\overline{E}$ at a distance r of 1.1 nm from the molecule on the dipole axis?	2.75
2.(a)	What is electric flux?	2
(b)	Obtain Coulomb's law from Gauss's law.	3.50
(c)	A point charge of 1.8 μC is at the centre of a spherical Gaussian surface of radius 55 cm. What is the net flux through the surface?	3.25
3.(a)	Calculate the total enclosed charge q for a parallel plate capacitor with dielectric using Gauss's law.	5
(b)	A parallel plate capacitor of plate area 11.5 cm^2 and plate separation 1.24 cm. A potential difference 85.5 V is applied between the plates. The battery is then disconnected, and a dielectric slab of thickness 0.78 cm and dielectric constant 2.61 is placed between the plates. What is the capacitance with the slab in place?	3.75
4.(a)	What is the current density and drift velocity of the charge carrier?	2
(b)	Establish the relationship between the current density and drift velocity.	4
(c)	What is the drift velocity of the conduction electrons in a copper wire of diameter 0.5 mm and length 20 m, when it is connected across the battery of 1.5V and the internal resistance 1.25Ω. (Here $\rho=8.96\times10^3\ kg/m^3$, $n=8.49\times10^{28}\ m^{-3}$)	2.75

Part B

5.(a)	State and explain Faraday's law of induction. Deduce its differential form.	3.75
(b)	Find the mutual inductance of two co-axial coils.	5
6.(a)	What do you mean by thermal emf?	2
(b)	What is Thomson effect?	3
(c)	Find an expression of the thermo-electric power.	3.75
7.(a)	Find expression for the growth and decay of charges of a capacitor through resistor with constant emf.	3.75
(b)	What is the time constant in an RC circuit? Show that the time constant has the unit of time.	2
(c)	In an RC circuit $R=0.4\times10^6\ ohm$ *and* $C=2.5\times10^6\ F$, in what time will the charge in the capacitor decay one fourth of its initial value?	3
8.(a)	An alternating emf is applied to a circuit consisting an inductor, capacitor and resistor in series. Obtain the expression for the current and impedance.	5.75
(b)	A circuit consists of a non-inductive resistance of 50Ω, an inductance of 0.3H and a resistance of 2Ω and a capacitor of 40μF in series and is supplied with 200 volts at 50 Hz. Find the impedance and the current in the circuit.	3

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engg.(CSE) 1st Year EVEN Semester 2016

Course: PHY 1221 (Applied Electricity and Magnetism)

Time: 3 Hrs. Full Marks: 52.5

[N.B. Answer SIX questions taking at least THREE from each part.]

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Part A

1.a) State superposition principle for electric charges. 1

b) Derive an expressions for the electric field at a point on the axial line due to an electric dipole. 4.75

c) An electric dipole of moment 2×10^{-8} cm is placed in a uniform field intensity $1.5\times10^{5}NC^{-1}$ (i) What maximum torque does the field exert on the dipole? (ii) How much work is done on turning the dipole end to end? 3

2.a) State and prove Gauss's law in electrostatics. 5

b) Charge is distributed uniformly over the surface of a sphere. Use Gauss's law to find the electric field at the points (i) Outside of the sphere (ii) On the surface of the sphere 3.75

3.a) Define capacitor. Classify the capacitors. Write down the major uses of capacitor. 1+1+1

b) Derive an expression for the energy stored by a charged capacitor. 3

c) The parallel plates of an air-filled capacitor are 1 cm apart. What will be the area A of each plate if capacitance is to be 0.25 μF? 2.75

4.a) Establish the relation between current density and electron drift velocity. 3.5

b) Deduce the condition for the balance of a Wheatstone's bridge. 3.5

c) Write short note about Superconductor. 1.75

Part B

5.a) State and prove Ampere's law. Apply it to calculate the magnetic field due to a solenoid. 3+3

b) Define self-inductance and mutual inductance. 2.75

6.a) Describe a moving coil galvanometer. Explain how current can be measured with it. 4

b) What are the differences between ballistic and deadbeat galvanometer? 2

c) Prove that the charge sensitivity of a B.G. is $2\pi/T$ times the current sensitivity. 2.75

7.a) Briefly describe about Seebeck and Peltier effects. 4

b) Write down the laws of addition of thermal electromotive forces. 4.75

8.a) A series circuit containing an inductor L and a resistor R is connected to a battery. Obtain the expression for the rate of decay of current. 4.75

b) An e.m.f. 10V is applied to a circuit having a resistance of 10Ω and in inductance of 0.5H. Find the time required by the current to attain 63.2% of its final value. What is the time constant of the circuit? 4

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-1, Even Semester Examination-2017
Course: PHY 1221 (Applied Electricity and Magnetism)
Time: 3 Hrs. Marks: 52.5
[Answer SIX (06) questions taking THREE (03) from each Section.]

Section A

1. (a) Define electric field, electric dipole, and dipole moment. 2
   (b) Find the potential energy of an electric dipole placed in a uniform external electric field. 3.75
   (c) A neutral water molecule in its vapor state has an electric dipole moment of magnitude 6.2×10^{-30} c.m. (i) How far apart are the molecules center's of positive and negative charge? (ii) If the molecule is placed in an electric field of 1.5×10^4 N/C, what maximum torque can the field exert on it? 3

2. (a) Prove that, for a point charge, Gauss' law is equivalent to Coulomb's law. 3.25
   (b) Apply Gauss' law to calculate the electric field for cylindrical symmetry. 3.5
   (c) Write down the few applications of Gauss' law. 2

3. (a) Define capacitor and capacitance. 3
   (b) Deduce the relation $C = \frac{\varepsilon_0 A}{d}$ for a parallel plate capacitor, where the symbols have their usual significance. 4
   (c) A storage capacitor on a Random Access Memory (RAM) chip has a capacitance of 55 μF. If the capacitor is charged to 5.3 V, how many excess electrons are on its negative plate? 1.75

4. (a) State Kirchhoff's laws of distribution in an electric network. 4
   (b) A metal rod of length 25 cm has resistance 7.5×10^{-6} Ω. The diameter of the rod is 0.40 cm. Calculate the resistivity of the metal. 2.75
   (c) Write short note about semiconductor. 2

Section B

5. (a) State and explain Faraday's law of electromagnetic induction. 3
   (b) Calculate the magnetic field inside a long straight wire with current. 3.5
   (c) A solenoid has length 1.23 m and inner diameter 3.55 cm and it carries a current 5.57 A. It consists of five closed packed layers, each with 850 turns along length. What is the B at its center? 2.25

6. (a) Write down the conditions for a moving coil galvanometer to be dead beat. 2
   (b) What is mutual inductance? Calculate the mutual inductance between two coaxial solenoids. 4.75
   (c) A solenoid of length 30 cm and area of cross-section 10 cm^2 has 1000 turns wound over a core of constant permeability 600. Another coil of 500 turns is wound over the same coil at its middle. Calculate the mutual inductance between them. 2

7. (a) What are thermocouple and thermal emf? 3
   (b) Define neutral temperature and temperature of inversion for thermocouple. 3
   (c) Calculate the maximum emf in a Fe-Pb thermocouple, the cold junction of which is kept at 0°C. Given a = 13.8 μ V/°C and b = -0.015 μV/(°C)^2. 2.75

8. (a) Define reactance and impedance of an AC circuit. 2
   (b) Analyze a series LCR circuit and explain the phenomenon of resonance. 4
   (c) Find the resonance frequency of an LCR series circuit for L = 10 mH, C = 0.02 μF and R = 20 Ω. 2.75

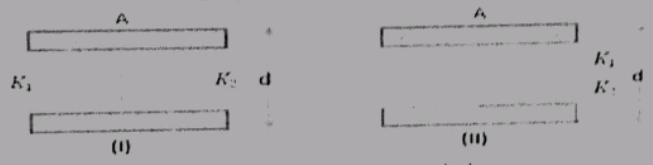
University of Rajshahi
Department of Computer Science and Engineering

B.Sc. (Engg.) Part-1, Even Semester, Examination-2018
Course: PHY-1221 (Applied Electricity and Magnetism)
Full Marks-52.5 Time: 3 hours

[N.B. Answer any six questions taking THREE from each of the Section]

Section-A

1. a) Define electric flux. 2
   b) Derive an expression for the capacitance of a spherical capacitor. 3.5
   c) The area of each plate of a parallel plate capacitor is A, and their separation is d. It is filled with two dielectrics of dielectric constants K_1 and K_2. Calculate the capacitance when the dielectrics are filled as in following figures. 3.25



2. a) State and prove Gauss's law in electro statistics. 5
   b) A long cylinder of radius a is uniformly charged with charge density λ per unit length. Find the field E at points inside and outside the cylinder. 3
   c) When do we apply Gauss's law? 0.75

3. a) What is dipole moment? Obtain an expression for the potential due to an electric dipole. 3.75
   b) Show that $\tau = \vec{P} \times \vec{E}$ for an electric dipole is placed in a uniform electric field $\vec{E}$. 2.5
   c) An electric dipole consist of two opposite charges of magnitude $q = 2.0 \times 10^{-6} C$ separated by 2.0cm. The dipole is placed in an external field of $q = 2.0 \times 10^{5}\ Nt/Coul$. Calculate the maximum torque on the dipole. 2.5

4. a) What is current density and drift velocity of a charge carrier? 2
   b) Establish the relation between drift velocity and current density. 3
   c) Calculate the drift velocity of free electron in copper from the following data: current density= $5 \times 10^{6}\ Am^{-2}$, density of copper= $9000 kg/m^3$, At. wt. of copper 64 gm/mole, Avogadro number = 6×10^{23} atoms/mole, electron charge = $1.6 \times 10^{-19} C$. 3.75

Section-B

5. a) State and prove Ampere's law. Apply it to calculate the magnetic field due to a solenoid. 6
   b) A solenoid is *1.0 m* long and *3.0cm* in mean diameter. It has *5* layers of windings of *850 turns* each and carries a current of *5A*. Calculate B at its center. 2.75

6. a) What are the Thomson effect and Peltier effect? 3
   b) Explain the thermoelectric diagrams. 1.75
   c) The *emf* of a thermocouple, one junction is kept at $0^o C$, is given by $E = at + bt^2$. Determine the neutral temperature, temperature of inversion, and the Peltier and Thomson coefficients. 4

7. a) Obtain an expression for the growth and decay of current in a *dc* circuit containing a resistance and inductance. 6.5
   b) An *emf 10 volts* is applied to a circuit having a resistance of *10 ohms* and an inductance of *0.5 Henry*. Find the time required by the current to attain 63.2% of its final value. 2.25

8. a) Obtain the expression for the mean value and the root mean square value of an alternating current. 3
   b) What is resonance frequency? Obtain an expression for it in case of LCR series circuit. Explain it for various R values. 3.75
   c) Find the resonance frequency of an LCR series circuit for $L = 10\ mH, C = 0.02\ \mu F$ and $R = 20\Omega$. 2

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. in Engineering 1st Year Even Semester Examination-2019

Course: **PHY-1221 [Applied Electricity and Magnetism]**

Marks: 52½ [Answer any **six** (06) questions taking **three** (03) from each section] Time: 03 Hours

Section-A

1.a)	What is electric dipole?	1
b)	How do you calculate the electric field intensity due to an electric dipole?	4¼
c)	An electric dipole consists of charges +2e and -2e separated by 0.87 nm. It is in an electric field of strength 4.4×10^6 N/C. Calculate the magnitude of the torque on the dipole when the dipole moment is (i) parallel to (ii) perpendicular to and (iii) antiparallel to the electric field.	3
2.a)	Write-down the few applications of Gauss law.	1¼
b)	Applying Gauss's law to calculate the electric field intensity due to a uniformly charged sphere (non-conducting) at points (i) outside the sphere, (ii) at the surface of the sphere, (iii) inside the sphere	5
c)	Draw the curve for the variation of the electric field intensity as function of radial distance due to a uniformly charged sphere (non-conducting) and discuss the results.	2½
3.a)	Define capacitance and a capacitor. On what factor does the capacitance depend?	1½+1½
b)	Find an expression for the capacitance of a parallel plate capacitor.	3
c)	A parallel plate capacitor consists of two square metal plates with 5.0 cm of side separated by 1.0 cm. A Sulphur slab of 6.0 mm thick and with $k = 4$ is placed on the lower plate, calculate the capacitance.	2¼
4.a)	Explain the following terms (i) insulator, (ii) semiconductor and (iii) conductor using the band-gap.	1½
b)	State and explain Kirchhoff's laws in an electric network. Apply these laws to find an expression for the current through the galvanometer in an unbalanced Wheatstone's bridge.	2+3¼
c)	The current in a simple series circuit is 5A. When an additional resistance of 2Ω is inserted, the current drops to 4A. What was the resistance of the original circuit?	2

Section-B

5.a)	State and explain Faraday's law of electromagnetic induction.	2
b)	What are the coefficients of self-inductance and mutual inductance? Find an expression for the self inductance of a solenoid.	2+3
c)	A 10 cm long solenoid having an air core has 100 turns and area of cross section 5 cm^2. Find the coefficient of self-inductance of the solenoid.	1¼
6.a)	Explain the Seebeck effect. Write down the names of thermos electrically negative and positive metals in thermos couple.	1¼
b)	How does Peltier effect differ from Joule heating effect?	2
c)	What are the natural temperature and inversion temperature?	2
d)	The emf (microvolts) in a lead-iron thermocouple, one junction of which is at 0°C is given by $E = 1874t - 3.4t^2$, where t is the temperature in degrees centigrade. Calculate the natural temperature, Peltier coefficient and Thomson coefficients.	3
7.a)	Derive an equation for the growth of charge in a dc circuit with inductance, capacitance and resistance in series.	6
b)	Using the above theory, explain the damped oscillatory case with figure.	2¼
8.a)	What is alternating emf?	1
b)	An alternating emf $E = E_0 \cos \omega t$ is applied to a circuit containing a resistance R and an inductance L in series. Find the current at any instant.	4¼
c)	Define the reactance and impedance of ac circuit.	1½
d)	In a series LR circuit, $E = 100\ volts, L = 25\ mh$ and $R = 50\Omega$. Calculate (i) the amplitude of the current, and (ii) the phase difference between the applied emf and current.	2

Full Marks: 52.5 CSE - PHY - 20222 Time: 3 Hours

[Answer any **Six** questions, Taking any **Three** from each section.
Use separate answer script for each section]

Section-A

1 a) State Gauss's law. Write down its integral and differential forms. 2.50

b) An insulating solid sphere of radius a has a uniform volume charge density ρ and carries a total positive charge Q. Using Gauss's law, Calculate the magnitude of the electric field at a point (i) outside the sphere (ii) inside the sphere. 4

c) A positive charge of 10^{-6} Coulomb is placed at the center of a spherical Gaussian surface of radius 0.5 m. Determine the flux through the surface of the sphere. 2.25

2. a) What is capacitor? Derive an expression for the capacitance of a parallel plate capacitor. 3.75

b) Derive the relation $\sigma_{ind} = \left(\frac{k-1}{k}\right)\sigma$ for a capacitor filled with a dielectric material, where symbols have their usual significances. 3

c) A parallel-plate capacitor has a plate separation d and plate area A. An uncharged metallic slab of thickness a is inserted midway between the plates. (i) Find the capacitance of the device. (ii) Show that the capacitance is unaffected if the metallic slab is infinitesimally thin. 2

3. a) Show that the energy stored per unit volume in a capacitor is $u_E = \frac{1}{2}\varepsilon_0 E^2$. 3

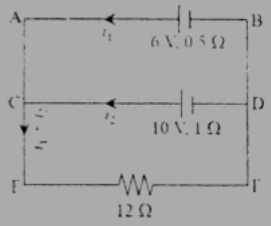
b) Prove that the volume polarization density ρ_p is equal to negative divergence of polarization $\vec{P}$. 3.25

c) The 12-gauge copper wire in a typical residential building has a cross-sectional area of $3.31 \times 10^{-6}\ m^2$. If it carries a current of 10.0 A, what is the drift speed of the electrons? Assume that each copper atom contributes one free electron to the current. The density of copper is $8.95\ g/cm^3$. 2.50

4. a) Explain how to distinguish conductor and semiconductor in terms of their conductivity variation. 3

b) What is the phenomenon of superconductivity? 2

c) State Kirchhoff's laws. Using Kirchhoff's laws, calculate the currents i_1 and i_2 in the following circuit. 3.75



Section-B

5. a) State and explain Faraday's law of electromagnetic induction. 3

b) What is motional e.m.f.? Derive an expression for the motional e.m.f. 3.25

c) Calculate the motional e.m.f. induced along a 20 km long conductor moving at an orbital speed of 7.80 km/s perpendicular to Earth's $5.00 \times 10^{-5}\ T$ magnetic field. 2.5

6. a) State Ampere's law. When do we apply Ampere's law? 1+2

b) A long straight wire carries a current of 20 A. An electron is travelling at 10^7 m/sec is 2 cm from the wire, what force acts on the electron if its motion is directed (i) towards the wire, (ii) parallel to the wire and (iii) at right angles to the direction given in (i) and (ii). 3

c) Distinguish ballistic and deadbeat galvanometer. 2.75

7. a) State the laws of addition of thermal electromotive forces. 2

b) Derive an expression for the total thermo e.m.f. of a thermocouple. 3.75

c) The thermo emf of an Al-Pb thermocouple, the cold junction of which is kept at 0 °C, is given by $E(\mu V) = 19.1\,t - 0.015\,t^2$, calculate the i) neutral temperature, ii) temperature of inversion, iii) Peltier and Thomson coefficient at 27 °C. 3

8. a) What is time constant? Mention its unit. 1.75

b) Show that for LR circuit the current I at time t during the growth is given by $I = I_0\left\{1 - e^{-\left(\frac{R}{L}\right)t}\right\}$. 4

hence determine the time constant for the circuit

University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science & Engineering
B.Sc. (Engineering) Part-I Even Semester Examination 2021
Course: PHY-1211 (Basic Electricity and Electrical Circuit)
Session: 2020-21
Full Marks: 60 Duration: 3(Three) Hours

Answer 06(Six) questions taking any 03(Three) questions from each part

Section-A

1. a) Explain the term i) Electric field, ii) Electric dipole, iii) Dipole moment. 3
   b) Derive an expression of potential energy of an electric dipole placed in a uniform external electric field. 4
   c) An electric dipole consists of two opposite charges of magnitude $2.0 \times 10^{-6}\ C$ separated by a distance $l = 1.0\ cm$. It is placed in an external electric field of $2.0 \times 10^{5}\ N/C$. What maximum torque does the field exert on the dipole. 3

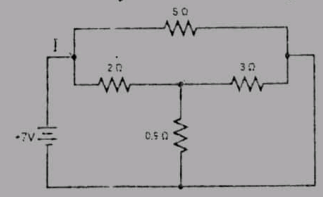
2. a) What do you mean by capacitor and capacitance? 3
   b) Derive an expression for the capacitance of a parallel plate capacitor. 4
   c) Calculate the capacitance of a parallel plate capacitor consists of two square metal plates with 5.0 cm of side separated by 1.0 cm, is placed in air media. 3

3. a) Apply the Kirchhoff's law to a wheat stone bridge and find the conditional for bridge balance. 4
   b) Write down the difference between semiconductors and superconductors. 3
   c) A metal rod of length 25 cm has resistance $7.5 \times 10^{-6}\ \Omega$. The diameter of the rod is 0.40 cm. Calculate the resistivity of the metal. 3

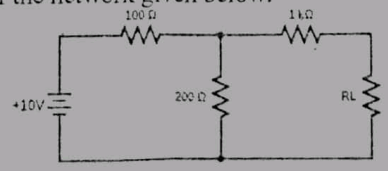
4. a) Find an expression for the self-inductance of a solenoid. 4
   b) A 10 cm long solenoid having an air core has 100 turns and area of cross-section 5 cm^2. Find the coefficient of self-inductance of the solenoid. 4
   c) State and explain Faraday's law of electromagnetic induction. 2

Section-B

5. a) State and explain Norton's theorem. 4
   b) Define the constant current source and a constant voltage source. 3
   c) Calculate the Current I supplied by the 7V battery in the following circuit. 3



6. a) State and prove the maximum power theorem for two terminal network. 4
   b) What is the Norton's equivalent of the network given below: 3



   c) State and explain Kirchhoff's current and voltage law. 3

7. a) Analyze a series LCR circuit and explain the phenomenon of resonance. 5
   b) Find the resonant frequency of an LCR series circuit for $L = 10\ mH, C = 0.02\mu F$, and $r = 20\Omega$. 3
   c) Write down the difference between steady current and transient current. 2

8. a) Define time constant with suitable example. 3
   b) A $5\mu F$ capacitor is discharged suddenly through a coil having an inductance of $2H$ and a resistance of 200Ω. The capacitor is initially charged to a voltage of $10V$. Find 3
      i. An expression for the current
      ii. The additional resistance required to give critical damping.
   c) What is double energy transients? Express the equation of loss free circuit when R=0. 4